

Motion Control #2: Impedance Control

Optimization and Learning for Robot Control

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1. Introduction to Interaction Control

Interaction control is essential when a robot must physically contact its environment or a human.

- **Direct Force Control:** Specifically regulates the contact force to a desired value ($f \rightarrow f_{ref}$). It requires explicit force feedback and often suffers from instability in stiff environments.
- **Indirect Force Control (Impedance):** Instead of commanding a force, we command the **dynamic behavior** (stiffness, damping, inertia) of the robot in response to external forces.

Key Concept: The robot behaves like a "programmable spring-damper" system.

2. Target Impedance Formulation

We want the closed-loop system to mimic a second-order mass-spring-damper system in Cartesian space:

$$M_d \ddot{x} + B \dot{x} + Kx = f_{\text{ext}}$$

where x is the displacement from a virtual equilibrium position.

Operational Space Projection: Starting from the joint-space dynamics $M\ddot{q} + h = \tau + J^T f_{\text{ext}}$, we project it into the task space:

$$\Lambda(q) \ddot{x} + \mu(q, \dot{q}) = J^{\#} \tau + f_{\text{ext}}$$

- $\Lambda(q) = (JM^{-1}J^T)^{-1}$: The Kinetic Energy Matrix in Operational Space.
- $\mu(q, \dot{q})$: The vector of Centrifugal, Coriolis, and Gravity forces in Task Space.

3. Deriving the Control Law

To achieve $M_d\ddot{x} + B\dot{x} + Kx = f_{ext}$, we solve for $\ddot{x}$ and substitute:

$$\tau = J^T [\Lambda M_d^{-1}(f_{ext} - Kx - B\dot{x}) + \mu - f_{ext}]$$

Special Case: Inertia Matching ($M_d = \Lambda$) If we do not attempt to change the robot's natural inertia (Λ), the control law simplifies significantly:

$$\tau = J^T (-Kx - B\dot{x} + \mu)$$

- **Significant Advantage:** In this case, **force sensor feedback is not required** to implement impedance control. The robot naturally provides the inertia Λ , and the control law adds the desired stiffness and damping.

4. Causality Analysis: Impedance vs. Admittance

Understanding which approach to use depends heavily on the robot hardware.

Impedance Control

- **Flow:** Motion $\rightarrow$ Force.
- **Implementation:** Torque-commanded.
- **Hardware:** Back-drivable robots (e.g., direct-drive, humanoids).
- **Pros:** Inherently stable in stiff environments.

Admittance Control

- **Flow:** Force $\rightarrow$ Motion.
- **Implementation:** Position-commanded.
- **Hardware:** Industrial robots with high-ratio gears.
- **Pros:** Good for soft environments and heavy-duty tasks.

5. Theoretical Insight: Passivity-based Stability

Why is impedance control stable even when contacting unknown environments?

Define a **Storage Function** (total energy) for the error system:

$$V = \frac{1}{2} \dot{x}^T \Lambda \dot{x} + \frac{1}{2} x^T K x$$

The time derivative $\dot{V}$, using the skew-symmetry property ($\dot{\Lambda} - 2C\Lambda = 0$), is:

$$\dot{V} = \dot{x}^T f_{\text{ext}} - \dot{x}^T B \dot{x}$$

Passivity Condition: If B is positive-definite ($B \succ 0$), then:

$$\int_0^T \dot{x}^T f_{\text{ext}} dt = V(T) - V(0) + \int_0^T \dot{x}^T B \dot{x} dt \geq -V(0)$$

The system is **passive** from f_{ext} to $\dot{x}$. This ensures that the robot can only dissipate energy, making the interaction with any passive environment (like a spring or a wall) stable.